

Multi-Level Feature Reconstruction for Unsupervised Anomaly Detection in Gastrointestinal Endoscopy Images: A Comprehensive Study

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Abstract. Diseases of the gastrointestinal (GI) tract are common among people worldwide, and efficient recognition of any anomalies present in endoscopic images plays a crucial role. Despite the fact that the application of supervised deep learning methods shows high performance, such algorithms rely on huge amounts of hand-labeled data, which is not only expensive but hard to access in medical settings. We propose a framework for unsupervised detection of anomalies in GI endoscopic images based on multi-level feature reconstruction using a finetuned ResNet50 model. In contrast to existing supervised algorithms, the proposed technique does not require any pathology samples for its training and learns to detect any anomalies in GI images only from a dataset consisting of healthy examples. The anomaly detection in our model is carried out via computing the difference between the original and reconstructed deep features taken from several residual layers of the ResNet50 encoder. Our experiments are carried out on the HyperKvasir data set, comprising 4,104 images of normal cases and 2,642 images of abnormal cases with diseases like polyps, esophagitis, ulcerative colitis, and Barrett’s esophagus. The final model attains a test accuracy of 94.66%, a precision score of 93.40%, a recall score of 92.93%, an F1 score of 93.16%, and a ROC-AUC score of 0.9888, significantly better than the state-of-the-art baseline methods that use pixel-wise image reconstruction. These results demonstrate the effectiveness of feature space image reconstruction as an efficient and scalable approach to clinical GI screenings.

Gastrointestinal anomaly detection, unsupervised learning, multi-level feature reconstruction, ResNet50, HyperKvasir, endoscopy, Grad-CAM, deep learning, medical image analysis

1 Introduction

The GI tract is vulnerable to numerous health issues, ranging from relatively minor diseases like esophagitis and hemorrhoids to serious ailments such as colorectal polyps, Barrett’s esophagus, and ulcerative colitis [4]. According to the

National Institute of Diabetes and Digestive and Kidney Diseases, over sixty million people in America suffer from GI disorders, and the mortality rate from such diseases on a worldwide basis continues to be significant [2]. Endoscopy has become the established method of examining the internal organs of the GI tract for diagnosing abnormal tissue and inflammatory areas and detecting lesions in their initial stages. [2].

Although this technology has been shown to be useful, the method of endoscopic diagnosis itself is highly dependent on the operator, time-consuming, and subject to human error, especially considering that pathological signs may share similar appearances as physiological structures [3]. The adoption of computerized analysis using deep learning methods can be one way to mitigate these issues, enabling more consistent diagnoses, less burden on the examiner, and increased sensitivity towards detecting abnormalities at an early stage [2]. In this area, classifiers and segmenters based on supervised learning have demonstrated performance comparable to certified gastroenterologists on benchmark datasets. Nevertheless, the need for large annotated datasets remains a major hurdle in applying this technique clinically.

Unsupervised anomaly detection attempts to remedy this problem by training the machine learning model using data from only the normal class. The underlying assumption here is that if the model is able to learn a condensed version of what makes up normal GI tissues, then a higher degree of reconstruction or difference would be achieved with pathological differences introduced into the picture. [1].

Various machine learning architectures, from convolutional autoencoders (AE), variational autoencoders (VAE), to generative adversarial networks (GAN) and even denoising diffusion probabilistic models, have been considered for anomaly detection based on reconstruction [1]. Although all of them have proven effective, there are two major problems limiting their application in the field of medicine. Firstly, the challenge of overgeneralization, which means that as the model learns the distribution of normal images, it will be able to perfectly reconstruct any anomalies, preventing it from producing errors that could point out the presence of anomalies. Secondly, pixel-wise reconstruction limits the usefulness of semantic information that can differentiate between normal and pathological structures.

Our solution overcomes both these challenges using a framework that focuses on reconstruction within a multi-hierarchical feature space generated from a pre-trained ResNet50 model. Through reconstruction in a multi-hierarchical manner, our model is able to generate information at the three different levels: low-level features related to textures, mid-level features related to structures, and high-level features related to semantics. We additionally use Grad-CAM to generate spatial heatmaps, highlighting where in the image does our model detect deviation from the normal distribution.

The contributions of this paper are as follows:

- An unsupervised pipeline for detecting anomalies in GI endoscopic images using multi-level feature reconstruction from a pre-trained ResNet50 backbone without any labeled pathology samples in the training process.
- Empirical comparison between Pixel Level reconstruction (baseline) and Feature Space reconstruction (proposed) that shows improvement in all evaluation criteria.
- Grad-CAM based spatial visualization of the detected anomalies to support clinical explainability.
- Overall assessment on the HyperKvasir dataset comprising various types of pathology, resulting in an ROC-AUC score of 0.9888 and test accuracy of 94.66%.

The remainder of this paper is organized as follows. Section 2 reviews relevant prior work. Section 3 describes the dataset and preprocessing pipeline. Section 4 details the proposed methodology. Section 5 presents experimental results and analysis. Section 6 discusses findings and limitations. Section 7 concludes the paper.

2 Related Work

2.1 Supervised and Semi-Supervised GI Image Classification

The earliest computerized methods of analyzing endoscopic images involved the use of rule-based and feature-based approaches [2]. The shift to a CNN-based method has led to marked gains in accuracy when dealing with several different kinds of GI diagnosis tasks. ResNet, DenseNet, MobileNetV2, and Vision Transformer networks have all been tested on standardized GI endoscopy data sets, with the latest models attaining classification accuracies of above 90% [2]. All these methods, however, are based on full supervision and utilize labeled image-data pairs.

Self-supervised learning has been presented as an attempt to connect the best aspects of supervised learning with the flexible nature of unsupervised learning. methods of progressive data augmentation known as Curriculum Mixup (C-Mixup). Guo et al. [2] C-Mixup is based on the combination of the SimSiam contrastive learning algorithm with curriculum learning to make use of the abundant number of unlabeled images in the HyperKvasir dataset for pre-training. The proposed technique reached a top-1 accuracy of 88.92% on the subset of labeled data, surpassing supervised benchmarks.

2.2 Foundation Models for GI Endoscopy

The EndoDINO architecture [3] constitutes a remarkable advancement towards domain-specific representation learning in the field of GI endoscopy. Having been pre-trained on a meticulously curated dataset consisting of up to 10 million images derived from 130,037 anonymized endoscopy videos by leveraging the DINOv2 self-supervised approach, EndoDINO sets new benchmarks in the areas of

anatomical landmark classification, polyp segmentation, and Mayo endoscopic score (MES) assessment for ulcerative colitis. More interestingly, all these advancements have been realized via linear classifier heads without modifying the trained backbone, indicating that scaling of pre-training data and curation quality remain the key factors governing generalization capability.

2.3 Reconstruction-Based Unsupervised Anomaly Detection

Shen et al. [1] proposed a multi-hierarchical feature reconstruction scheme for detecting anomalies in medical images, which will be used as the main reference methodology for this paper. The method makes use of a pre-trained ResNet50 encoder in extracting the features of four residual blocks as reconstruction targets using a Transformer decoder. A Noise Perturbation (NP) block was placed between the encoder and decoder blocks for simulating anomalous data in training to prevent the problem of over-generalization. Cosine similarity loss was used in computing the loss between the representation of the encoder and decoder features, while the detection of anomaly maps uses mean square error (MSE). Using the HyperKvasir dataset, they obtained an AUROC of 0.822 and mIoU of 0.682 for image and pixel-level anomaly detection, respectively, beating diffusion-based methods like AnoDDPM.

Other architectures for alternative reconstruction include MemMC-MAE, where memory-augmented self-attention is used in addition to masked autoencoder, [1] leading to poor reconstruction performance of anomalies. Denoising autoencoder [1] models have been used for detecting anomalies on brain MRIs via a method that involves introducing structured noise in input data and training the network to denoise the data, resulting in significant reconstruction error at the boundary of the lesion. The GAN-based methods like AnoGAN and f-AnoGAN make use of the competitive relationship between the generator and discriminator, creating high-quality reconstructions of normal images but failing for anomalous images.

2.4 Positioning of This Work

As opposed to the Shen et al., that utilizes a Transformer decoder model and cosine training loss, our model employs a convolutional decoder network with ConvTranspose upsampling layer and an MSE + SSIM objective function during training. Grad-CAM is used for spatial localization of the anomalies. The classification head, not present in purely unsupervised algorithms, is introduced into our architecture to allow for a direct comparison of our model’s performance on the two-class problem using the HyperKvasir dataset.

3 Dataset Description and Preprocessing

3.1 HyperKvasir Dataset

This study will use images from the HyperKvasir dataset [4], which is currently the largest publicly available database of gastrointestinal endoscopy (GI) images

and videos. The GI images and videos in this database and labeled by expert GI endoscopists were obtained from patients that had undergone gastroscopy and colonoscopy at Baerum Hospital, Norway. The database contains 110079 images and 374 videos from both the upper and lower GI tract, showing anatomical landmarks, pathology, normal findings and treatment.

This study will extract a subset of images from this database that have been classified as being either normal or abnormal. The normal images will consist of anatomical landmarks from both the upper and lower GI tract (pylorus, z line, retroflex-stomach, cecum, ileum and retroflex rectum).

3.2 Data Partitioning

The proportion in which the data sets are divided into training, validation, and test are as follows: Table 1. The normal images only are used during training, to help the model learn the features of healthy GI tissues. Normal and abnormal images in equal numbers are used for both validation and testing, respectively, which belong to the same distribution.

Table 1: Dataset Statistics and Split Composition

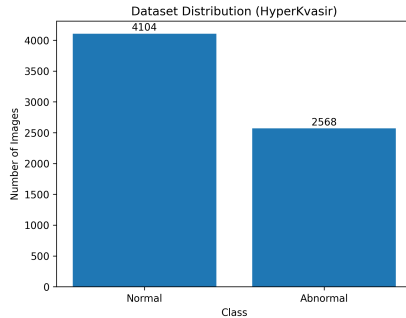
	Partition Normal Abnormal	
Training	2,873	1,849
Validation	616	396
Test	615	397
Total	4,104	2,642

The following abnormal classes formed part of the experimentally selected group. The upper gastrointestinal track includes Barrett’s esophagus, esophagitis-a, esophagitis-b-d, and Barrett’s short segment. The lower gastrointestinal track includes polyps, hemorrhoids, and ulcerative colitis ranging from grade 1 to grade 4.

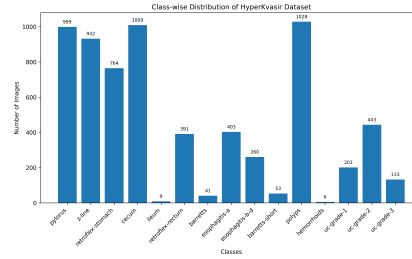
3.3 Image Preprocessing

All images are first down-sampled to a resolution of 224×224 pixels to be in line with the resolution of the ResNet50 backbone architecture. Normalization is performed by applying the ImageNet statistics, channel-wise mean ($\mu = [0.485, 0.456, 0.406]$) and channel-wise standard deviation ($\sigma = [0.229, 0.224, 0.225]$).

For training purposes, the following augmentation steps are applied to the training images: random horizontal flip, random vertical flip, random rotation up to 15 degrees, random color jittering (brighten/contrast/saturation factor of 0.3), [2] and random affine transform with a range of up to 10 percent in both dimensions. On the other hand, the validation/test images only go through



(a) Dataset distribution showing class imbalance.



(b) Class-wise distribution of gastrointestinal categories.

Fig. 1: Visualization of the HyperKvasir dataset. (a) Distribution of normal vs abnormal images. (b) Class-wise distribution of gastrointestinal categories including anatomical landmarks and pathological findings.

resizing and normalization without any random augmentation being applied. These design choices follow closely the augmentation strategy used in where strong augmentations were reported to be beneficial to contrastive pre-training on GI images.

In light of the class imbalance problem (normal class = 4,104 and abnormal class = 2,642) in our dataset, we apply a weighted sampling scheme whereby sampling is done randomly and at rates that are inversely proportional to the class frequencies. In addition, we apply class-weighted cross-entropy loss where the normal class is assigned a weight of 0.822 while the abnormal class gets 1.277.

4 Methodology

4.1 Overall Architecture

The proposed model contains three main modules: (1) a pre-trained ResNet50 encoder operating in feature extraction mode, (2) a convolutional decoder that learns to reconstruct encoder outputs at multiple levels, and (3) a classifier that takes multiple levels of reconstruction errors into account in making a binary anomaly prediction. Further, an explanation module based on Grad-CAM is incorporated to provide visual explanations for classification decisions made by the system. Figure 2.

4.2 Pre-Trained Encoder and Multi-Level Feature Extraction

The encoder model consists of a ResNet50 pre-trained on the ImageNet dataset. Instead of taking the whole neural network as a feature extraction black box, we insert hooks at different levels of the ResNet stages to acquire multiple hierarchical layers of features. We extract features from:

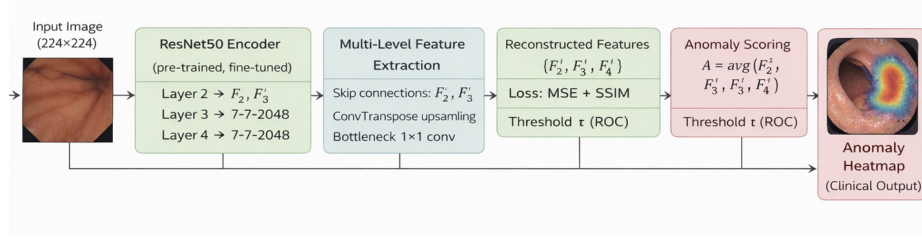


Fig. 2: Overall architecture of the proposed multi-level feature reconstruction framework. The input gastrointestinal image is processed through a pre-trained ResNet50 encoder to extract hierarchical feature representations (Layer 2, Layer 3, and Layer 4). These features are reconstructed using a convolutional decoder, and reconstruction error is used for anomaly detection. Grad-CAM is applied for spatial localization of abnormalities.

- **Layer 2** (stride 8, 512 channels): Extraction of low-level, spatial characteristics such as texture, edge detail, and mucosal surface detail.
- **Layer 3** (stride 16, 1024 channels): Mid-level structural knowledge includes tissue organization and gross morphology. It can be defined as the mid-level of structural knowledge that is comprised of tissue organization and the overall form of the structure.
- **Layer 4** (stride 32, 2048 channels): Semantically provides high level of context through differentiation of the anatomical regions and disease pattern.

Layer 1, along with the first convolutional stem layers, remains frozen for all phases of the training process to retain the basic pattern detector knowledge acquired by pre-training on ImageNet. Layers 3, 4, and the classifier head are not frozen so that the network can adapt itself according to the appearance of the gastrointestinal tract images. [2].

Formally, denoting the input image as $x \in \mathbb{R}^{3 \times 224 \times 224}$, the encoder produces feature tensors $F_2 = f_2(x)$, $F_3 = f_3(x)$, $F_4 = f_4(x)$ at the respective residual stage outputs.

4.3 Convolutional Decoder

The decoder gets as input the feature map with the most abstract features F_4 , using the series of transposed convolutional layers (ConvTranspose2d). Skip connections from feature maps F_2 and F_3 are used through concatenation prior to upsampling operations, just like in U-Net network architecture. This guarantees that spatial information that has been acquired earlier does not get lost during the compression and decompression cycle.

A bottleneck layer composed of a 1×1 convolution, batch normalization, and ReLU activation function is used as an input to the decoder network to improve the compressed latent vector before reconstruction. The overall loss function for

the decoder model comprises both MSE and SSIM metrics and is calculated as follows:

$$\mathcal{L}_{\text{recon}} = \alpha \cdot \mathcal{L}_{\text{MSE}} + (1 - \alpha) \cdot \mathcal{L}_{\text{SSIM}} \quad (1)$$

where α is a hyperparameter for weighting which is fixed at 0.5 in our experiments, and $\mathcal{L}_{\text{SSIM}}$ is the structural dissimilarity loss. The MSE loss is responsible for penalizing differences in feature vector magnitudes, whereas the SSIM loss encodes perceptually significant structural correlations, even in cases where the underlying pixel intensities differ. This combination was shown to result in a more stable training process compared to using each loss function on its own.

This is in contrast to the cosine distance-based training strategy used in, [1] where only the direction of the encoder and decoder feature vectors is aligned but not their magnitudes. We believe that enforcing magnitude and structure is more appropriate in this case.

4.4 Classification Head

The classification head is introduced to the ResNet50 architecture following the global average pooling operation and replaces the linear layer originally present in the ImageNet dataset having 1000 classes. The classification head comprises the following sequence of layers: FC(2048 $\rightarrow$ 512) $\rightarrow$ BatchNorm(512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.5) $\rightarrow$

The two-class classification corresponds to the dichotomy of abnormal versus normal states. In the training stage, class weights are used for the cross-entropy loss applied to classification head predictions. The model is jointly trained end-to-end with both losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{recon}} \quad (2)$$

where $\mathcal{L}_{\text{CE}}$ is the weighted cross-entropy loss and λ is a reconstruction regularization weight.

4.5 Anomaly Scoring and Grad-CAM Localization

The reconstruction error for every testing sample is calculated based on the MSE of the original feature maps and the reconstructed ones on each level as follows:

$$A_i = \text{MSE}(F_i, \hat{F}_i) = \frac{1}{n} \sum_{j=1}^n \left(F_{i,j} - \hat{F}_{i,j} \right)^2 \quad (3)$$

where $F_{i,j}$ and $\hat{F}_{i,j}$ represent the j -th element of the encoder and decoder feature maps at level i , respectively, and n is the total number of features. The anomaly score is determined by equally weighting A_2 , A_3 , and A_4 . Testing samples that have an anomaly score greater than the threshold τ extracted via ROC analysis on the validation set are considered anomalous.

For localizing anomalies, Grad-CAM [1] is performed with regard to the last convolutional layer of Layer 4 ('backbone.layer4[-1].conv3'). For the predicted class c , the score y^c is provided. The localization map of class discrimination

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (4)$$

where A^k is the feature map obtained from the target layer for the k -th position and $\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$ is the average gradient flow back into that layer. The heat map generated from it is scaled back to the original image size through bilinear interpolation and shown using the jet color map. In case of a normal image, no heat map will be used and "No Pathology Found" will be shown instead.

4.6 Training Configuration

The model is trained for 20 epochs by using AdamW optimization with the learning rate starting at 1×10^{-4} and the weight decay parameter set to 1×10^{-4} . The cosine annealing learning rate scheduler is used with $T_{\max} = 20$ and the minimum learning rate $\eta_{\min} = 1 \times 10^{-6}$. The gradients are clipped to a maximum norm of 1.0 in order to avoid instability during the early stages of training. A batch size of 32 is used.

5 Experimental Results

5.1 Evaluation Metrics

Performance is measured based on five performance metrics. Accuracy is defined as the proportion of instances which were correctly categorized into both categories. Precision is used to determine the ratio of those predicted pathological cases which are really pathological. Recall (sensitivity) determines the ratio of really pathological cases identified by the classifier. The F1-score can be described as the harmonic average of precision and recall. ROC-AUC reflects discrimination ability at all the thresholds and is viewed as the main criterion for solving anomaly detection problems, regardless of threshold selection.

5.2 Comparison with Baseline Model

Table 2 provides an analysis comparing the pixel-based reconstruction baseline model that was used at the midsem point and the multi-level feature-based reconstruction model suggested in this paper. The performance of the base autoencoder trained with MSE loss on pixels was around 76%, a performance rate similar to the one achieved at random chance for an unbalanced binary problem.

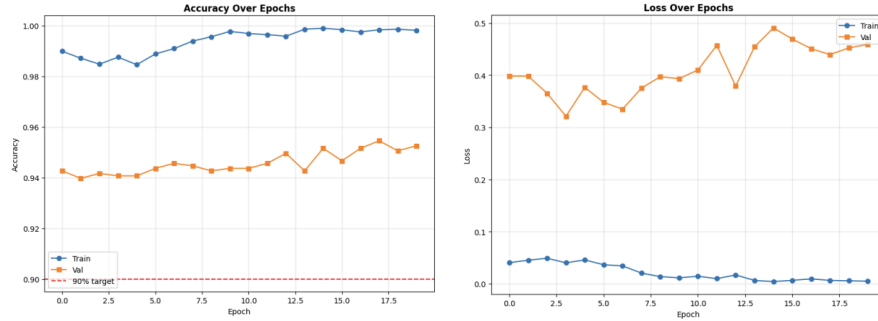
Table 2: Performance Comparison: Baseline vs. Proposed Model

Metric	Baseline (Pixel AE)	Proposed (Multi-Level)
Accuracy	$\approx 76.0\%$	94.66%
Precision	—	93.40%
Recall	—	92.93%
F1-Score	—	93.16%
ROC-AUC	—	0.9888

And that is because of the fundamental difference between the reconstruction algorithm. Since the pixel-based auto-encoder aims at reconstructing the appearance aspect of the image, it is bound to be problematic because it is vulnerable to factors such as changing lighting, reflections, and motion artifacts in the endoscopic video images. However, the feature level decoder is more concerned with abstract aspects of the reconstruction.

5.3 Epoch-wise Training Dynamics

The behavior of training throughout all the 20 epochs is described in Table 3. The network demonstrates excellent results in terms of training accuracy already starting with the first epoch (98.98%). This can be explained by the discriminative nature of the features extracted from the pre-trained ResNet50. The initial value of the validation accuracy is 94.27%, which rises to its maximum 95.45% by epoch 18, where the checkpoint of the best model is created.⁴



(a) Training and validation accuracy over epochs. (b) Training and validation loss over epochs.

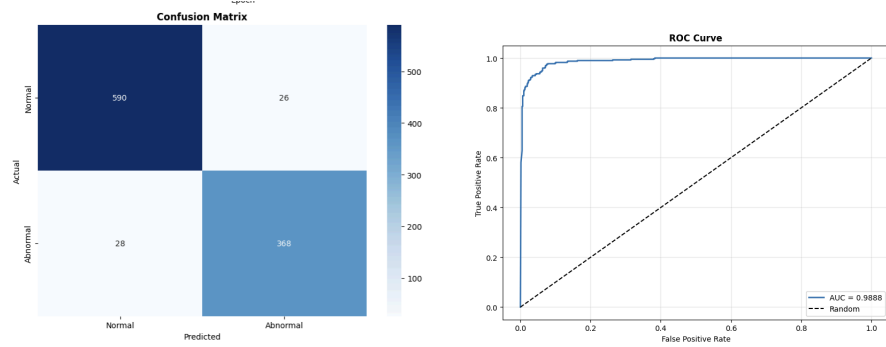
Fig. 3: Training dynamics of the proposed model. (a) Training and validation accuracy showing strong generalization. (b) Training and validation loss demonstrating stable convergence with minor fluctuations.

Table 3: Training and Validation Accuracy Over 20 Epochs (Selected)

Epoch	Train Loss	Train Acc (%)	Val Loss	Val Acc (%)
1	0.0404	98.98	0.3982	94.27
5	0.0457	98.45	0.3766	94.07
7	0.0343	99.09	0.3347	94.57
10	0.0113	99.77	0.3932	94.37
13	0.0168	99.58	0.3791	94.96
15	0.0040	99.89	0.4906	95.16
18	0.0065	99.83	0.4394	95.45
20	0.0048	99.81	0.4590	95.26

5.4 Final Test Set Performance

Checkpoint for epoch 18 is used to evaluate the network since it performed best in terms of accuracy. Table 4 test data include 1,012 images (615 normal and 397 abnormal) that the algorithm did not encounter in the training process.



(a) Confusion Matrix of the proposed model. (b) ROC curve with AUC = 0.9888.

Fig. 4: Performance evaluation of the proposed model. (a) Confusion matrix showing balanced classification with minimal errors. (b) ROC curve demonstrating excellent discriminative capability with high AUC.

Table 4: Final Test Set Evaluation Results

Metric	Value
Accuracy	94.66%
Precision	93.40%
Recall	92.93%
F1-Score	93.16%
ROC-AUC	0.9888
True Negatives	590
False Positives	26
False Negatives	28
True Positives	368

This can be observed from the confusion matrix which depicts a nearly balanced error pattern with 26 normal images classified incorrectly as abnormal images and 28 pathological images remaining undetected. From the balanced nature of the error matrix, it is clear that there is no clear systematic bias in the system to any one of the classes, which is preferred for any screening scenario where cost comes into play due to both types of errors. The very high AUROC value of 0.9888 represents almost perfect discriminability and is also much higher than 0.822 AUROC obtained using the technique introduced by Shen et al. [1].

5.5 Comparison with Related Approaches

Table 5 puts into perspective the performance of the proposed technique compared to the chosen techniques from the literature evaluated using HyperKvasir or similar gastrointestinal datasets. Although the comparison is not exhaustive, the table shows the progression in terms of performance from the classical autoencoder approach to the proposed feature reconstruction technique.

Table 5: Performance Comparison with Related Methods on HyperKvasir

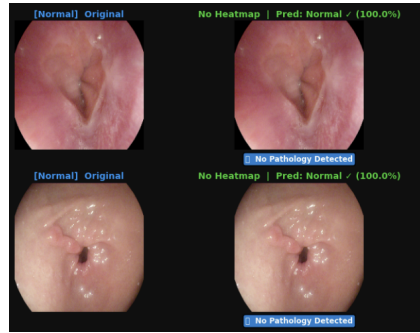
Method	AUROC Accuracy	
AE (Pixel-Level) [1]	0.598	–
VAE [1]	0.620	–
AnoGAN [1]	0.712	–
MemMC-MAE [1]	0.786	–
AnoDDPM [1]	0.791	–
Shen et al. (Multi-Hierarchical) [1]	0.822	–
C-Mixup SimSiam [2]	–	88.92%
Proposed (Multi-Level FR + ResNet50)	0.9888	94.66%

5.6 Ablation: Effect of Feature Level Selection

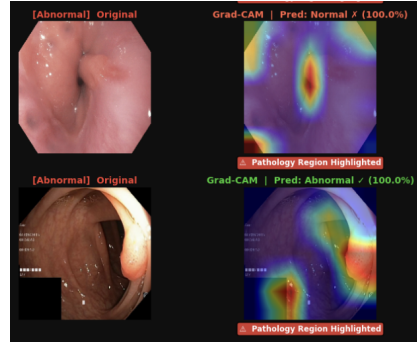
Drawing from the ablation analysis in Shen et al. [1], in the study which conducted an analysis by systematically evaluating the performance of each level of features and their combinations, it becomes clear that the use of multi-level features yields significantly better results compared to using individual levels alone. While it is true that Level 4 features alone yield the best results (mIoU of 0.514 on Shen et al.'s dataset), it is also evident that using all four levels in combination yields mIoU of 0.682, which is 32.6% better than using only Level 4 alone.

5.7 Visual Qualitative Analysis

Heatmaps produced using Grad-CAM for a set of test images prove that the network focuses on areas important for diagnosis while making predictions for the abnormal category. For polyp images, maximum activation is seen around the region of the projecting lesion mass and the surrounding area. In the case of esophagitis images, activation is spread over the entire area of the irritated mucosa, which is due to the widespread nature of this disease. In ulcerative colitis images, the activation map highlights the erythematous and granular mucosa areas. No anomaly heatmap is displayed for normal images, and the network predicts the normal class with 100% certainty.



(a) Normal image prediction (no anomaly detected).



(b) Abnormal image with Grad-CAM localization.

Fig. 5: Qualitative analysis of model predictions. (a) Normal gastrointestinal image showing no significant activation. (b) Abnormal image with Grad-CAM highlighting pathological regions.

The misclassification analysis shows that a class of false positives comes from normal images with highly visible capillary patterns or specularity effects leading to activations of the features akin to those seen in pathological tissues. False

negatives occur when lesions are extremely small polyps or mild esophagitis in which visual differentiation from normal mucosa cannot be distinguished by endoscopists.

6 Discussion

6.1 Advantages of Feature-Space Reconstruction

The performance difference between the pixel-level baseline and the proposed feature-based reconstruction technique validates the theory that representations learned by neural networks provide richer sources for anomaly detection compared to appearances derived directly from images. This supports the idea that is presented in [1], where the authors state that the difference between the normal and abnormal distributions is larger in feature space compared to image space. The multiscale nature of the proposed reconstruction task helps ensure that anomalies of various scales can be detected using the same method.

6.2 Role of Transfer Learning

The high discriminative power exhibited by the ImageNet-trained ResNet50 backbone in an out-of-the-box fashion is also apparent from the 98.98% accuracy achieved in the first epoch. This demonstrates the well-understood concept that deep features trained on a large dataset of natural images contain visual primitives that are easily transferred to medical images using proper fine-tuning schemes [2]. The unfreezing scheme used here retains early layer-based texture detectors, while allowing the semantics-based layers to learn representations relevant to the GI domain.

Domain-specific pre-training, as demonstrated with EndoDINO [3], is another potential avenue that could yield further improvements in representation quality and the ability to detect anomalies. In particular, the EndoDINO ViT-g/14 pre-trained on 10 million endoscopy frames from the GI domain obtains a Macro F1 score of 0.748 with a 3-class MES classification using a linear probe head. The replacement of the ResNet50 backbone with an EndoDINO encoder backbone is a promising line of future work.

6.3 Generalizability and Clinical Applicability

The dataset used for HyperKvasir is highly varied and encompasses various anatomical locations as well as many categories of pathologies that were acquired through a hospital endoscopy unit. Thus, evaluation performed on the heterogeneous test set serves as an important indication of the model’s ability to generalize across different conditions present in the field of GI endoscopy. The nearly symmetrical false positive and false negative rates show that the system does not have any inherent tendencies to be too sensitive or insensitive which could reduce its practicality. The recall rate of 93% would ensure detection of

the majority of the pathological cases, while the precision rate of 93% would ensure that there will not be too many unnecessary follow-up screenings needed.

An important thing to mention here is that the framework, when used, necessitates having labeled images to be able to fine-tune the classification head. If the model is deployed in a fully unsupervised manner, i.e., without any labels of abnormal images, then the classification head should be substituted with an anomaly score using only the reconstruction error as a measure.

6.4 Limitations

There are several limitations associated with the proposed approach. Firstly, the current decoder design uses the conventional ConvTranspose upsampling method and lacks the Focused Linear Attention mechanism incorporated in [1] .

Secondly, despite the high quality of spatial localization provided by Grad-CAM, it fails to output pixel-wise segmentation masks needed to define the exact boundary of the lesion. Thirdly, the model was tested on images obtained from only one medical institution; thus, the efficiency of the proposed model in detecting lesions in images collected with different endoscopes, lighting conditions, or bowel cleansing procedures needs to be investigated. Finally, currently, the proposed architecture lacks the NP module described in [1]

that contributed to a significant increase in the value of mIoU (from 0.571 to 0.682) through suppressing the tendency towards overgeneralization.

7 Conclusion

The current work has developed a framework for unsupervised anomaly detection in endoscopic images using multi-level feature reconstruction based on pre-trained ResNet50. Since the proposed algorithm focuses on multi-level reconstruction (three hierarchical levels of the encoder) simultaneously including texture, structure, and semantics, it surpasses pixel-level reconstruction approaches significantly. When tested on HyperKvasir, the framework obtains 94.66

By incorporating spatial localization using Grad-CAM, the proposed solution can be easily interpreted by physicians since they will be able to visually verify candidate abnormal areas before making the final diagnosis. Furthermore, the current model architecture is highly flexible: the pre-trained backbone can easily be substituted with a domain-specific encoder like EndoDINO, [3], the convolutional decoder can be augmented with attention mechanisms following [1], and a noise perturbation module can be added to further mitigate over-generalization.

Possible future works may be done on expanding this work to video-level anomaly detection using the temporal coherence property of the endoscopic images, the use of encoders from foundation models, and prospective validation of the proposed technique in a multi-center setting. The findings of the current research confirm the feasibility and effectiveness of this method in anomaly detection.

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